

Teaching Portfolio — Kimball Germane

Narrative Summary of the Teaching Portfolio for Rank & Status Review

Kimball Germane · Computer Science · 15 August 2026 · CFS Review

The most fundamental way I have grown as an instructor is that I no longer adjust my courses solely by feel: I gather and respond to *feedback* during the semester and adjust the course each summer in light of all of the in- and post-semester feedback. A manifestation of this change is the teaching reports and goals themselves: whereas the early entries (circa 2021) are based on feel, the later are based more on student and grading feedback.

Three examples of this practice:

1. I have run a mid-course evaluation survey in each of my last seven offerings of CS 330 (Fall 2022 through Winter 2026), which allows me to respond to student feedback the same semester it is given.
2. I place the goals for a semester, distilled from feedback, in that semester's entry, so that the later teaching report can be compared directly against the goals, tightening the connection between feedback and practice.
3. The nature of my formal verification course (CS 401R, now CS 430) allows me to mechanically check how many proof obligations (the fundamental problem form) were satisfied relative to attempted and assigned.

This practice has allowed me to see and in some cases quantify my growth. For example, the mid-course evaluation survey shows the rate of students who would describe lectures as “impenetrable” (word given to them) decline from a peak of 61% in Winter 2023 to 7% (1 of 15 respondents) in Winter 2026. (At the same time, the rate of students that described lectures as “valuable” remained high.) This trend is concurrent with the adoption of new pedagogical techniques (see below).

I have also grown by studying and adopting the pedagogical techniques of others grounded in the evidence-based learning literature. Addressing feedback that students did not feel completely comfortable asking questions, I instituted pedagogical techniques to remove structural impediments. For example, I explicitly instructed students to ask questions not only if they needed clarification, but if they could sense that anyone might need clarification, which decreases the exposure of anyone asking a question. I particularly like such policies because they genuinely increase the amount of learning that occurs and the sense of

belonging that students feel while it does.

Over the course of my record keeping, my view of what experience my course could offer students has expanded. While my courses have always incorporated prayer, beginning Fall 2022 I incorporated a few once-a-week practices not directly related to the course material. For example, one of these practices is “lifhack Thursday” in which the floor is open to share or report on academic, spiritual, and general lifehacks. Through the mid-course evaluation survey, many students—unprompted—attested to the value of these practices and several students corroborated this sentiment out-of-band.

I adjust the course substantially each summer and feedback allows me to detect issues. For instance, students reported a higher workload across 2025, due, I believe, to some new course tooling. As part of the adjustment this summer, I am reworking the tooling in response to some of the specific feedback students have given.

I adjust in response to student feedback and also to the field moving, most markedly with the rise of coding agents. I have asked students about their use of coding assistants since Fall 2022, added course content relevant to that use, such as type systems, wrote an explicit AI policy in Fall 2025, and moved exams from take-home to in-class in Winter 2026. The graduate seminar ran on programming languages and coding agents this winter.

Teaching Journal

The goals for a particular semester of a course are recorded next to the report for that course and labeled by the semester which produced them.

Beginning Fall 2025, CS 430 is the officialized version of CS 401R; journal entries use the number in effect at the time.

Standing practices

- Every class opens with a prayer (since Fall 2020)
- Students are encouraged to ask questions not only if they need clarification but if they can sense or imagine that others might need clarification (since Fall 2020)
- Once a week I share a brain teaser that breaks the ice and provokes a good mindset (since Fall 2022)
- Once a week I open the floor for academic, spiritual, and general lifehacks (“Lifhack Thursday”) (since Winter 2023)
- Students introduce themselves to each other each class before they discuss problems (since Fall 2023)
- Students hold up the number of fingers corresponding to the number of non-clarifying responses they’ve given (since Fall 2023)
- I ask questions until there are no more answers and not until the correct answer is given (since Fall 2023)

Across my seven surveyed offerings of CS 330, 24 respondents have named brain teasers and 17 have named Lifhack Thursday as positive aspects of the course that should be kept.

(These were totally unprompted—the survey doesn’t name or ask about either.) I have had several students communicate privately later that a particular lifehack has been extremely valuable to them.

Winter 2027

CS 601R

Goals From Winter 2026

Student Learning

I expect that, by the next time I run this course, there will have been significant shifts in the practice and usage of coding agents that will bear on programming languages theories and models in interesting ways. As a consequence, I believe that running the course with the same theme would yield something meaningfully different.

Learning Environment

I received solid feedback about the learning environment. I will maintain the structure and tone of discussions.

Fall 2026

CS 330

Goals From Winter 2026

Student Learning

The thrust of the course has to change. It can no longer be about how programming languages are the interface for a developer; instead, it must be about how the particular features and constraints of a given programming language affect the kinds of guarantees one can claim about agent-generated code.

Learning Environment

The class setting is going to look more like a lab in the sense that students will be oriented and then run experiments. An obvious failure mode is an experimental setup that permits no agency nor any meaningful prediction, but I think that will be avoided by the fact that there are many different ways to express specifications and, for any given specification, write code that meets them.

Processes of Improvement

I am updating the tooling of the course in accordance with the change in thrust. Specifically, the students will use a new language with familiar syntax and semantics but that has the warts of existing languages smoothed off. Those warts will then be used as case studies to illustrate considerations that must be made in the “real world” when applying the techniques that they learn in a more idealized setting. The design, which allows the language to be

interpreted with static and dynamic constraints, is settled; most of its implementation is complete as well.

I need to make sure I encourage completing the course surveys and student ratings.

CS 430

Goals From Fall 2025

Student Learning

We must do more exercises in class, but they must still read the material beforehand. They need multiple, spaced exposures to the proof concepts and expert guidance to learn how to apply them.

Learning Environment

Rather than demonstrate a proof outright, I think I will have each student come to the front and “drive” while the class talks them through an exercise. It’s not a perfect solution, but it does provide students with a bit of cover when they get stuck.

Winter 2026

CS 330

Goals From Fall 2025

Student Learning

Students are not as able as I expect to organize the information in each unit into a mental framework. I will provide for each unit a scaffolding that reflects the hierarchy and relationships between concepts.

Learning Environment

I will provide more opportunities to work in partnerships and groups, but also ensure that they are sufficiently primed for any task I give them.

Teaching Report In this run of the course (in part due to the existence of AI tools), exams that were previously take-home were administered in class.

Student Learning

1. What went well? I added additional “cheat sheets” for each unit and called their attention to existing ones to provide a scaffolding for concepts. I adjusted the course offering to provide more content on type systems, which is especially relevant in the current environment. The lowest fraction (7% of 15 respondents) of students described lectures in the mid-course survey as “impenetrable”.
2. What could be improved? This run of the course came in the wake of Claude Code, which was a specific and clear example of agent programming where the user isn’t

required to look at the code. The existence of this example significantly damaged the premise of the class that, as software developers, programming languages would be their primary communication medium of software construction.

Learning Environment

1. What went well? I had many more exercises and activities where class members talked to each other and that seemed to regularly break the ice among the students.
2. What could be improved? Whether or not it was due to the existence of Claude Code or coding-agent and AI trends more generally, I had many more students not come to class than in previous semesters. I had fairly low turnout of student ratings (48%) relative to past semesters.

CS 601R

Teaching Report This run of the course was rocked similarly to 330. Consequently, this semester was devoted to topics and readings at the intersection of programming languages and LLM-based coding agents. When this plan was presented to students on the first day of class, they lamented requiring relevance to coding agents would be too limiting. This fear was unfounded, fortunately, and the set of allowable topics remained satisfyingly broad. Moreover, the LLM context grounded class discussion in a particular and fruitful way.

Student Learning

1. What went well? We covered a lot of interesting and cutting edge concepts.
2. What could be improved? It would be good (if possible) to nail down all material up front rather than having it driven by student interest.

Learning Environment

1. What went well? The engagement and daily class discussions were excellent.
2. What could be improved? Sometimes the discussions become a bit ungrounded and it would be good to maintain a grounding mechanism.

Fall 2025

CS 330

Goals From Fall 2024

1. Provide students more and more-consistent opportunities to work problems in class, alone, with a partner, in groups, and as an entire class. Iterate collaboration at these various levels to shepherd students at different levels of understanding through the same content.
2. Unify the two approaches to programming languages the class has offered historically—interpreters (implementer-perspective) and concepts (user-perspective)—by having students gradually modify the interpreters to support the standard model of programming languages seen in the other perspective. Present an understanding of this unified view as a main course objective.

3. Systematically and incidentally connect the course topic to their regular programming practice, and facilitate this by using a less-alien language (which some student ratings comments mentioned).

Teaching Report An improvement that I attempted for fall semester was to cleanly separate the syntax of the implementing and implemented languages, which helps students see the act of interpretation more clearly. There were some hiccups that made this a bit of a bumpy experience, mostly due to providing them the right tooling. With the right tooling in place, I am confident about the approach for the coming semesters.

I have vacillated between letting the students choose their own language and standardizing on a language that forces them to learn a new way of programming. Unfortunately, each has their advantages and distinct disadvantages. Moving forward, I expect to maintain a uniform language in the class since (i) it gives the students enough exposure to internalize the new way, and (ii) it allows for a more systematic and deep grading process, since I can make comments about code organization that will apply generally.

I also introduced an explicit AI policy that affirms the usefulness of these tools but restricts their use in the course, on the grounds that using them well requires a base of domain knowledge that the course exists to build.

I encouraged the class members to fill out the intro survey and mid-course evaluation survey and student ratings.

Student Learning

1. What went well? I attempted to unify the two approaches to programming languages the class has offered historically. The vast majority (95% of 19 students who opted to take it) of students aced the final. (It may look like there is a selection effect going on here, but many of the students who didn't take it didn't need to to get the grade they wanted.) This large majority, of course, could mean the final was too easy—and it may have been—but it did at least moderately test knowledge that a vast majority of students lacked coming in.
2. What could be improved? I dedicated some time in the class on review that otherwise would have gone toward delivering the punchline and was unable to give them an assignment where they made the connection themselves. Mid-course responses describing the lectures as “impenetrable” rose to 27% of 22 respondents, from 15% of 26 in the winter offering.

Learning Environment

1. What went well? I was able to provide students more and more-consistent opportunities to work problems in class, alone, with a partner, in groups, and as an entire class, especially at the beginning of the semester.
2. What could be improved? The practice needs to be consistent through the end of the semester. Students reporting the workload as “about what is reasonable for a 3-credit course” fell to 54% of 26 respondents in winter and 50% of 22 in fall, from 85% of 33 in Winter 2024; I attribute this trend in part to a bit of friction of the course tooling.

CS 430

Goals From Fall 2024 (taught as CS 401R)

1. Motivate and foreshadow concepts before they are encountered in just reading or exercises
2. More opportunities for feedback about proof style and technique in assignments and in class

Teaching Report Student Learning

1. What went well? Students closed 88% (611 of 694) of assigned proof obligations and 53% (1412 of 2665) of *unassigned* proof obligations. (The nature of the assignments allows me to mechanically check, and proving things in this system is really fun.)
2. What could be improved? I need to make the foreshadowing deliberate to ensure that it happens.

Learning Environment

1. What went well? I had fewer and lower requirements to come to the front of the class.
2. What could be improved? Unfortunately, I didn't recognize until late how much of a bead this gave me on the progress of individual students.

Professional Development

I read Shriram Krishnamurthi's "Pedagogy Recommendations" in December 2025. He recommends many books, several of them because they are explicitly evidence-based, and I looked at several of them:

- The ABCs of How We Learn
- Understanding How We Learn: a Visual Guide
- The Teaching Gap
- Scienceblind

This post also reviews some of the techniques that he uses and that I adopted after observing his teaching a few years ago:

- think-pair-share
- finger counts when raising hands
- taking answers until students have no more to give and not merely taking answers until they're correct

For CS 430 I also participated and presented in a seminar offered by the math department on using the Lean tool to formalize mathematics. Lean is very similar to Rocq, the tool used in CS 430. Many students participated as well in the seminar with a range of math and CS backgrounds.

Winter 2025

CS 330

Teaching Report I updated my intro survey to ask more comprehensive questions about their use of LLMs and AI assistants to code.

Student Learning

1. What went well? Mid-course survey: 15% of 26 respondents described the lectures as “impenetrable”—the lowest so far.
2. What could be improved? Students reporting the workload as “about what is reasonable for a 3-credit course” fell to 54%, from 85% in Winter 2024.

Fall 2024

CS 330

Teaching Report One of my PhD students, Tim Whiting, taught the course. I helped him make substantial changes to the course, including those that introduced students to particular languages at key points in the course progression.

CS 401R

Goals From Fall 2023

1. More opportunities for motivated students to go deeper with particular topics
2. More feedback about proof style and technique in assignments

Teaching Report I achieved both goals in 401R. I included more background reading as an out-of-class activity and assigned reflective, open-ended exercises as homework. In class, we reviewed many of these exercises. Having had the background exposure outside of class, we were able to go deeper on those topics, and thereby achieved the first goal. Additionally, students presented their solutions regularly at the beginning of class and we had discussions about their approaches and techniques, thereby achieving the second goal.

Student Learning

1. What went well? Students went from essentially zero formal verification experience to proving non-trivial properties. The nature of the course allows me to mechanically check what fraction of proof obligation instances were closed: 70% of 3,989. One of the final projects also mechanized a paper from my research area (*Abstracting Abstract Machines*).
2. What could be improved? Classroom discussion could introduce proof techniques earlier so that students have more than the textbook explanation before applying them in exercises

Learning Environment

1. What went well? Had students come up to the front to complete exercises and present solutions which provided good feedback
2. What could be improved? Some students may have been embarrassed going up to the front, and it would be good to provide a feedback channel that was less public

Professional Development

I read research and retrospectives on the flipped classroom model to implement it in my PhD student's run of 330. I applied the model in 401R by moving baseline material out of class so that we had more time/bandwidth for more open-ended exercises.

Winter 2024

CS 330

Goals From Fall 2023

1. Convey the foundational framework more explicitly
2. Maintain the pattern of in-class exercises on which students can work together
3. Introduce students to new languages and features more systematically

Teaching Report I succeeded in the first two goals. To achieve the first, I oriented a thread of the course around a standard model of languages. To achieve the second, I planned in-class exercises to do regularly throughout the semester and administered them.

Student Learning

1. What went well? Based on their performance, many of the students understand programming language concepts from an implementation and user perspective. According to the mid-course evaluation survey, 91% of students (of 33 respondents) found lectures "valuable". The share describing them as "impenetrable" fell to 21%, from 61% of 38 the year before.
2. What could be improved? A legible course purpose/theme. Based on student ratings comments, many students didn't understand the purpose of the class nor perceived any overarching theme

Learning Environment

1. What went well? A wide variety and significant fraction of students asked questions and contributed, which I encouraged.
2. What could be improved? The amount of scaffolding that students are given. According to student ratings comments, when asking students to work in partnerships or groups, many felt lost and then, seeing their partner/group was lost too, frustrated.

Fall 2023

CS 330

Goals From Fall 2022

1. More in-class exercises
2. More explicit models (in both senses)
3. More concrete connections to “mainstream PL”

Teaching Report The textbook I use for 330 had a substantially-revised edition released at the beginning of 2023. I adjusted 330 over the summer to use this edition.

I believe I met the goals I set out to:

1. The class had significantly more in-class exercises that students worked on individually, in small groups, and as a class.
2. The class introduced a standard model of programming languages that acted as a through line in the course to discuss various language features and implementation techniques.
3. Finally, the course had many more in-class exercises and outside-class assignments that had students explore topics relative to concrete programming practice.

I feel that the adjustments I made neglected to offer a foundational framework that students can use to analyze languages and features. As a result, students had more experience and knowledge about a variety of features, but do not have the intellectual tools to analyze their interactions or analyze novel features.

Student Learning

1. What went well? The vast majority of students became fluent in functional programming. According to the mid-course evaluation survey, only 36% of 25 respondents found lectures “impenetrable”, compared to 61% of 38 in winter 2023.
2. What could be improved? The foundational framework could be conveyed more explicitly and integrated more into assignments. The course should expand students’ experience with programming languages more than it does, in addition to unifying their programming experience (gleaned from a mid-course evaluation survey).

Learning Environment

1. What went well? A wide variety and significant fraction of students asked questions and contributed, which I encouraged. Several (six) students also explicitly mentioned lifehacks and brain teasers as being valuable in the mid-course evaluation survey.
2. What could be improved? In the latter half of the semester, I did not have as many in-class exercises that students could work on in groups.

CS 401R

Goals From Fall 2022

1. More opportunities throughout semester to formulate properties
2. Need to give students smaller venues in class (like groups or partnerships) to voice questions and solidify understanding

Teaching Report The textbook I use for this course is superb, and I saw the phenomenon of students working ahead more prominently this semester. This phenomenon complicates lecture because some students are working through the material for the first time, while others are far ahead. I definitely do not want to discourage more engagement with the material. I think a good response might be to create supplementary material for each unit that students can use to dive deeper without forging ahead. I mechanically checked that four students closed 72% of 1,829 proof-obligation instances.

Professional Development

I worked with Shriram Krishnamurthi to gain access to his course for his new textbook and studied his practices as an instructor. He incorporates several practices that I will adopt, such as:

- having students talk to each other about the answer to a question before opening up the floor
- having students introduce themselves to each other before chatting with each other
- getting answers until there are no more answers instead of answers until the correct one is given
- having students hold up the number of fingers corresponding to how many times they have made a comment that day

I also derived material from his textbook and assignments and worked with him on shared course infrastructure.

Winter 2023

CS 330

Teaching Report I introduced a new practice where, once a week, I open the floor for students to share academic, spiritual, and general lifehacks. I offer nominal extra credit to students who use this opportunity to report on lifehacks they have tried and to students whose lifehacks were tried. I got good feedback about this and will continue the practice.

CS 601R

Goals From Winter 2022

Have students implement PL artifacts to gain deeper understanding and connection with their regular programming. Twofold purpose: (1) use own model as framework for new knowledge, validated by implementation; (2) allow students to speak from experience and appeal to authority less.

Teaching Report This class is organized around discussions of PL papers where students take turns leading the discussion. As many students who take this class do not have a PL background, it is easy for the discussion lead to have no more insight or grasp than other class members. To remedy this, I required discussion leads to implement an artifact from the

paper before the discussion which they could then demonstrate to add more context to the discussion. My perception is that discussion leads this semester had a much firmer grasp on the substance of the paper and were able to speak more from practice to address student questions.

Fall 2022

CS 330

Goals From Fall 2021

My goal for 330 is to make a class that students love and feel is relevant. I intend to do this by reorganizing the course to have smaller, more regular assignments which are more focused and directed than previously.

Teaching Report I reorganized 330 as described and it appeared to be much more manageable for students across the board. I need to draw connections more explicitly so that students (across the board) can see the relevance. I reworked my intro survey and ask about use of GitHub Copilot. I incorporated a mid-course evaluation survey that I will run each semester to be able to respond to feedback within the same semester. I started to do brain teasers once a week for a variety of reasons.

Student Learning

1. What went well? The vast majority of students became fluent in functional programming. The mid-course evaluation survey revealed that 44% of 27 respondents considered the lectures “impenetrable.” (The word was a given option.)
2. What could be improved? Some of the assignments are not apparently connected to their regular programming practice. The mid-course evaluation survey revealed that 44% of 27 respondents considered the lectures “impenetrable.”

Learning Environment

1. What went well? A broad swath of students were willing to ask questions in class.
2. What could be improved? Some lectures didn’t require students to take an active role.

CS 401R

Goals From Fall 2021

Give the students more opportunities to formalize software properties.

Teaching Report I gave 401R students more opportunities to formalize software properties by creating some lab-type lectures. It went well.

Student Learning

1. What went well? Students went from essentially zero formal verification experience to proving non-trivial properties Four students closed 75% of 1,708 proof-obligation instances.
2. What could be improved? Students need more opportunities to formulate properties throughout the course.

Learning Environment

1. What went well? Had many lab-style lectures which built skills efficiently.
2. What could be improved? Some students afraid of asking questions because they thought their classmates understood.

Winter 2022

CS 601R

Goals From Winter 2021

My goal for 601R is to prepare PL grad students to do research and non-PL grad students to access the PL literature in their own research.

Teaching Report My 601R goal was ill-formed because I did not/could not observe the students' ability to access the PL literature in their research.

Student Learning

1. What went well? Students were exposed to a wide variety of cutting edge PL topics
2. What could be improved? Students need a better framework to organize and arrange the knowledge from this exposure

Learning Environment

1. What went well? Lots of discussion and engagement.
2. What could be improved? Some students w/o PL background were intimidated by others with, and the class should foster their willingness to speak up.

Fall 2021

CS 330

Goals From Fall 2020

Streamline the experience for students. My reviews indicate (to me) that I need to find better ways to explain certain concepts and a better way to convey the course organization. I have made explicit changes in the course to address each and look forward to seeing the effect.

Teaching Report I attempted to organize 330 around a large interpreter project, but shifted to a traditional course midway through the semester.

CS 401R

Goals From Fall 2020

Publicize it effectively to get more students. I think my 330 courses will offer a bit of a pipeline, but I would like to reach out more broadly too.

Teaching Report I had a higher initial enrollment for my 401R class but several students dropped early on. I followed up with them. Some students dropped because they felt unprepared (not having taken 330), and two dropped because they didn't need the credit. Seven students finished the term, closing 64% of attempted proof obligations.